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Travel control device for hybrid vehicle

Abstract

A travel control device for a hybrid vehicle includes: an acceleration control part for controlling a traveling motor generator or an engine based on a depression amount of an accelerator pedal; and a regenerative control part for controlling the traveling motor generator with a regenerative control intensity set by an operation part. The acceleration control part includes: an EV priority control portion for prioritizing EV traveling over engine traveling. The regenerative control part includes: a restriction portion for restricting or prohibiting regenerative braking; and a forced regeneration portion for performing regenerative braking on a driving wheel and rotationally driving a power generation motor generator by electricity generated in the traveling motor generator to rotationally drive an engine, if the regenerative braking intensity is increased by the operation part in a state where the restriction portion restricts or prohibits the regenerative braking. The acceleration control part deselects an EV priority mode if the regenerative braking intensity is increased by the operation part in a state where the EV priority mode is selected and acceptance of electricity to the driving battery is restricted.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a travel control device for a hybrid vehicle.

BACKGROUND

(2) Patent Document 1 discloses a travel control device for a hybrid vehicle, which includes: a traveling motor generator configured to power-drive a driving wheel of the vehicle and configured to perform regenerative braking on the driving wheel; a driving battery configured to supply electricity to the traveling motor generator if the traveling motor generator power-drives the driving wheel and configured to recover electricity from the traveling motor generator if the traveling motor generator performs regenerative braking on the driving wheel; a power generation motor generator configured to generate electricity to charge the driving battery; and an engine configured to rotationally drive the power generation motor generator and be rotationally driven by the power generation motor generator. The hybrid vehicle includes an operation part for setting a regenerative braking intensity of the traveling motor generator, whereas the travel control device includes a regenerative control part configured to control the traveling motor generator such that the traveling motor generator performs regenerative braking on the driving wheel with the regenerative control intensity set by the operation part. The regenerative control part includes: a restriction portion configured to control the traveling motor generator such that regenerative braking is restricted or prohibited if the driving battery restricts acceptance of electricity; and a forced regeneration portion configured to control the traveling motor generator such that the traveling motor generator performs regenerative braking on the driving wheel and configured to rotationally drive the power generation motor generator by electricity generated in the traveling motor generator to rotationally drive an engine if the regenerative braking intensity is increased by an control lever in a state where the restriction portion controls the traveling motor generator such that regenerative braking is restricted or prohibited.

(3) Further, the hybrid vehicle is known which includes an EV switch for selecting an EV priority mode, and prioritizes, if the EV priority mode is selected, EV traveling where the rotational drive of the engine is stopped over engine traveling where the engine is rotationally driven.

CITATION LIST

Patent Literature

(4) Patent Document 1: JP6500368B

SUMMARY

Technical Problem

(5) Meanwhile, in a case where the hybrid vehicle disclosed in Patent Document 1 is provided with the EV switch and the EV priority mode is selected, if the regenerative braking intensity is increased by the operation part in a state where the driving battery restricts acceptance of electricity to the driving battery even though EV traveling is prioritized over engine traveling, the engine is rotationally driven by rotationally driving the power generation motor generator with electricity generated in the traveling motor generator. Therefore, it is necessary to deselect the EV priority mode.

(6) The present disclosure has been made in view of the above-described problems, and an object of the present disclosure is to provide a travel control device for a hybrid vehicle capable of deselecting an EV priority mode if a regenerative braking intensity is increased by an operation part in a state where a driving battery restricts acceptance of electricity to the driving battery, even though the EV priority mode is selected.

Solution to Problem

(7) In order to achieve the above object, a travel control device for a hybrid vehicle according to the present disclosure, includes: a traveling motor generator configured to power-driving a driving wheel of the vehicle and configured to perform regenerative braking on the driving wheel: a driving battery configured to supply electricity to the traveling motor generator if the traveling motor generator power-drives the driving wheel and configured to recover electricity from the traveling motor generator if the traveling motor generator performs regenerative braking on the driving wheel; a power generation motor generator configured to generate electricity to charge the driving battery; and an engine configured to rotationally drive the power generation motor

generator and be rotationally driven by the power generation motor generator. The hybrid vehicle includes: an EV switch for selecting an EV priority mode; and an operation part for setting a regenerative braking intensity of the traveling motor generator. The travel control device includes: an acceleration control part configured to control rotational drive of the traveling motor generator or rotational drive of the engine such that the driving wheel is power-driven based on a depression amount of an accelerator pedal; and a regenerative control part configured to control the traveling motor generator such that the traveling motor generator performs regenerative braking on the driving wheel with the regenerative control intensity set by the operation part. The acceleration control part includes: an EV priority control portion configured to prioritize EV traveling where the rotational drive of the engine is stopped over engine traveling where the engine is rotationally driven, if the EV priority mode is selected. The regenerative control part includes: a restriction portion configured to control the traveling motor generator such that the regenerative braking is restricted or prohibited, if the driving battery restricts acceptance of electricity to the driving battery; and a forced regeneration portion configured to control the traveling motor generator such that the traveling motor generator performs regenerative braking on the driving wheel and configured to rotationally drive the power generation motor generator by electricity generated in the traveling motor generator to rotationally drive the engine, if the regenerative braking intensity is increased by the operation part in a state where the restriction portion controls the traveling motor generator such that the regenerative braking is restricted or prohibited. The acceleration control part deselects the EV priority mode if the regenerative braking intensity is increased by the operation part in a state where the EV priority mode is selected and the acceptance of electricity to the driving battery is restricted.

Advantageous Effects

(8) With a travel control device for a hybrid vehicle according to the present disclosure, an EV priority mode can be canceled if a regenerative braking intensity is increased by an operation part in a state where a driving battery restricts acceptance of electricity to the driving battery, even though the EV priority mode is selected.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a diagram schematically showing the configuration of a hybrid vehicle equipped with a travel control device according to Embodiment 1.
- (2) FIG. 2 is a diagram schematically showing the configuration of a driving battery shown in FIG. 1.
- (3) FIG. 3 is a view schematically showing a shift lever shown in FIG. 1.
- (4) FIG. 4 is a view schematically showing a paddle switch shown in FIG. 1.
- (5) FIG. 5 is a diagram for describing a relationship between a shift position and an operation of the shift lever or the paddle switch.
- (6) FIG. 6 is a view schematically showing a multi-information display.
- (7) FIG. 7 is a flowchart showing the contents of travel control of the travel control device according to Embodiment 1.
- (8) FIG. 8 is a flowchart, showing the contents of travel control of the travel control device according to Embodiment 2.

DETAILED DESCRIPTION

(9) A travel control device for a hybrid vehicle according to embodiments will be described below with reference to the accompanying drawings. It is intended, however, that unless particularly specified, dimensions, materials, shapes, relative positions and the like of components described in the embodiment or shown in the drawings shall be interpreted as illustrative only and not intended

to limit the scope of the present invention. The hybrid vehicle described here is a plug-in hybrid vehicle (PHEV, PHV), but it is not limited to this and may be a hybrid (HV) vehicle.

Embodiment 1

(10) [Configuration of Hybrid Vehicle 1]

(11) As shown in FIG. 1, a hybrid vehicle **1** of Embodiment 1 is a hybrid vehicle powered by a traveling motor generator **21**, a power generation motor generator **23**, and an engine **25**, and includes a power drive unit (PDU) **3**, an engine control unit (ECU) **4**, and a driving battery **5**, in addition to the traveling motor generator **21**, the power generation motor generator **23**, and the engine **25**.

(12) The traveling motor generator **21** is a motor generator configured to power-drive a driving wheel **27** of the vehicle and configured to perform regenerative braking on the driving wheel **27**, and the traveling motor generator **21** is controlled by the power drive unit **3**. Electricity is supplied to the traveling motor generator **21** from the driving battery **5** connected to the power drive unit **3** if the driving wheel **27** is power-driven, and electricity (electricity regenerated in the traveling motor generator **21**) is recovered from the traveling motor generator **21** to the driving battery **5** via the power drive unit **3** if the driving wheel **27** is subjected to regenerative braking. A transaxle **29** is disposed between the traveling motor generator **21** and the driving wheel **27**. The transaxle **29** is a device integrating a transmission, a final gear, and a differential gear. Power of the traveling motor generator **21** is transmitted to the driving wheel **27** via the transaxle **29** during vehicle acceleration, and braking power of the traveling motor generator **21** during vehicle deceleration is transmitted to the driving wheel **27** via the transaxle **29**. Whereby, traveling is possible in which the traveling motor generator **21** is rotationally driven in a state where the rotational drive of the engine **25** is stopped. In the following description, “traveling in which the traveling motor generator **21** is rotationally driven in the state where the rotational drive of the engine **25** is stopped” is referred to as “EV traveling” and being in a state of EV traveling is referred to as an “EV traveling mode”.

(13) The power generation motor generator **23** is a motor generator configured to generate electricity to charge the driving battery **5**, and the power generation motor generator **23** is controlled by the power drive unit **3**. Electricity generated by the power generation motor generator **23** is charged to the driving battery **5** connected to the power drive unit **3**. In the hybrid vehicle **1** according to embodiment 1, electricity from the driving battery **5** alone may not be sufficient to produce a maximum output if the vehicle accelerates rapidly, so electricity generated by the power generation motor generator **23** is added to electricity from the driving battery **5** and supplied to the traveling motor generator **21**. Whereby, traveling while starting the engine **25** and supplying electricity generated by the power generation motor generator **23** to the driving battery **5** or the traveling motor generator **21** is possible, and electricity generated by the power generation motor generator **23** is added to electricity supplied from the driving battery **5** if electricity generated by the power generation motor generator **23** is supplied to the traveling motor generator **21**. In the following description, “traveling while starting the engine **25** and supplying electricity generated by the power generation motor generator **23** to the driving battery **5** or the traveling motor generator **21**” is referred to as “series traveling” and being in a state of series traveling is referred to as a “series traveling mode”.

(14) The engine **25** is an engine configured to rotationally drive the power generation motor generator **23** and be rotationally driven by the power generation motor generator **23**, and the engine **25** is controlled by the engine control unit **4**. For example, an output shaft (crank shaft) of the engine **25** is connected to an output shaft of the power generation motor generator **23** directly or indirectly via a gear train or the like. Thus, the power generation motor generator **23** is rotationally driven by rotationally driving the engine **25**, and conversely, the engine **25** is rotationally driven by rotationally driving the power generation motor generator **23**. Consequently, if the power generation motor generator **23** is rotationally driven by electricity generated (regenerated) in the traveling motor generator **21**, the engine **25** is also rotationally driven. In this case, the engine **25**

becomes a load, and electricity generated (regenerated) in the traveling motor generator **21** is consumed by the power generation motor generator **23**.

(15) The engine **25** is, for example, a gasoline engine, but it is not limited to the gasoline engine and may be an internal combustion engine such as a diesel engine. In the hybrid vehicle **1** according to Embodiment 1, the engine **25** is coupled to the power generation motor generator **23** via the above-described transaxle **29**, the motor generator is rotationally driven if the engine **25** is subjected to a combustion operation, and the engine **25** is rotationally driven if the power generation motor generator **23** is rotationally driven. In the hybrid vehicle **1** according to Embodiment 1, a clutch **291** is disposed in the transaxle **29**, and the power of the engine **25** is transmitted to the driving wheel **27** via the transaxle **29** by connecting the clutch **291**. Whereby, traveling while transmitting the power of the engine **25** to the driving wheel **27** is possible by starting the engine **25** as well as connecting the clutch **291**, and the power of the engine **25** is added to the power of the traveling motor generator **21**. In the following description, “traveling while transmitting the power of the engine **25** to the driving wheel **27**, by starting the engine **25** as well as connecting the clutch **291**” is referred to as “parallel traveling” and being in a state of parallel traveling is referred to as a “parallel traveling mode”. Electricity generated by the power generation motor generator **23** in parallel traveling is charged to the driving battery **5** connected to the power drive unit **3**.

(16) As shown in FIG. 2, the driving battery **5** is formed by a battery pack **50** in which a plurality of (many) battery cells **51** are housed in a single container. The battery cells **51** housed in the battery pack **50** are grouped into several groups, and the plurality of battery cells **51** included in the group form one battery module **52**. The driving battery **5** thus configured is provided with a current sensor **53**, a leakage sensor **55**, a cell monitor unit (CMU) **57**, and a battery management unit (BMU) **59**. The current sensor **53**, the leakage sensor **55**, and the cell monitor unit **57** can manage (monitor) a voltage and a temperature of the battery cells **51** included in the group. The battery management unit **59** can manage the battery module **52** and can individually manage all the battery cells **51** through the cell monitor unit **57** incorporated in each group. Whereby, the battery management unit **59** can monitor the voltage and the temperature of each battery cell **51**, in addition to being able to estimate the capacity and state of charge (SOC) of the battery pack **50**, and can determine whether electricity can be accepted to the driving battery **5**, in addition to being able to diagnose a failure. Thus, the battery management unit **59** constitutes a determination part for determining whether the acceptance of electricity to the driving battery **5** is restricted.

(17) Whether electricity can be accepted to the driving battery **5** is determined, for example, by determining full charge, a high battery temperature, and a low battery temperature. The full charge is determined by estimating the capacity and state of charge of the battery pack **50**, and it is determined that the battery is fully charged if predetermined state of charge (threshold) is exceeded. The high battery temperature is determined by the temperature of the battery cell **51** having the highest temperature (highest cell temperature) among the plurality of battery cells **51**, and it is determined that the battery is hot if the highest cell temperature exceeds a predetermined temperature (threshold). The low battery temperature is determined by the temperature of the battery cell **51** having the lowest temperature (lowest cell temperature) among the plurality of battery cells **51**, and it is determined that the battery is cold if the lowest cell temperature is less than a predetermined temperature (threshold). Then, if it is determined that the battery is either of fully charged, hot, or cold, the battery management unit **59** restricts the acceptance of electricity to the driving battery **5**. On the other hand, if all the determinations that the battery is fully charged, hot, or cold are resolved in the state where the acceptance of electricity to the driving battery **5** is restricted, the restriction on accepting electricity to the driving battery **5** is lifted. Restricting the acceptance of the driving battery **5** to electricity includes a case where the amount of electricity accepted to the driving battery **5** is restricted, besides a case where the acceptance of electricity to the driving battery **5** is prohibited.

- (18) The hybrid vehicle **1** according to Embodiment 1 described above is controlled by a travel control device (EVECU) **6**. The travel control device **6** is composed of, for example, an input/output interface (not shown), besides a processor (not shown) composed of a computing device, a register for storing an instruction or information, a peripheral circuit, etc., and a memory (not shown) such as ROM (Read Only Memory) or RAM (Random Access Memory).
- (19) The travel control device **6** is connected to an accelerator pedal sensor **71**, a brake pedal sensor **73**, a vehicle speed sensor **75**, and an operation part **77**.
- (20) The accelerator pedal sensor **71** is a sensor for detecting a depression amount of an accelerator pedal, and the brake pedal sensor **73** is a sensor for detecting a depression amount of a brake pedal. The vehicle speed sensor **75** is a sensor for detecting a vehicle speed. The vehicle speed sensor **75** is constituted by, for example, a wheel speed sensor, and the vehicle speed is obtained by converting a wheel speed detected by the wheel speed sensor.
- (21) The operation part **77** is a part for setting a regenerative braking intensity of the traveling motor generator **21**. As shown in FIGS. **3** and **4**, the operation part **77** according to Embodiment 1 is composed of a selector lever **773** disposed on a center console **771** and a regeneration level selector **777** disposed in the vicinity of a steering wheel **775**, but the operation part **77** is not limited to this, and may be composed of, for example, various operation devices such as an operation lever or an operation button that can be operated by a user seated in a driver's seat.
- (22) The selector lever **773** and the regeneration level selector **777** are operation devices for Changing shift positions, and are configured such that a regeneration level is set for each shift position. The regeneration level is set in association with the regenerative braking intensity, and the regenerative braking intensity is changed by changing the regeneration level. As shown in FIG. **5**, shift positions that can be set by the selector lever **773** and the regeneration level selector **777** are six shift positions of, for example, from B0 (regenerative braking intensity: low) to B5 (regenerative braking intensity: high).
- (23) As shown in FIG. **3**, the selector lever **773** is of a joystick type and can be operated to respective positions of R (reverse), N (neutral), D (drive), and B (regenerative brake) from a home position, and the basic shift position D (B2) is selected by operating the selector lever **773** to D. Further, a shift position "B3" with one step higher regenerative braking intensity is selected by operating the selector lever **773** to B once, and a shift position "B5" with maximum regenerative braking intensity is selected by operating the selector lever **773** to B again. Thus, the regenerative braking intensity is increased by operating the selector lever **773** to B once, and the regenerative braking intensity is decreased by operating the selector lever **773** to D.
- (24) As shown in FIG. **4**, the regeneration level selector **777** is of a paddle type, and is formed by a negative-side selector **7771** and a positive-side selector **7773**. The negative-side selector **7771** is used to displace to a shift position with one step higher regenerative braking intensity, and each operation of the negative-side selector **7771** displaces to a shift position with one step higher regenerative braking intensity. The positive-side selector **7773** is used to displace to a shift position with one step lower regenerative braking intensity, and each operation of the positive-side selector **7773** displaces to a shift position with one step lower regenerative braking intensity. Thus, the regenerative braking intensity is increased by operating the negative-side selector **7771**, and the regenerative braking intensity is decreased by operating the positive-side selector **7773**.
- (25) As shown in FIG. **1**, the travel control device **6** according to Embodiment 1 described above is connected to an EV switch **79** for selecting an EV priority mode. As shown in FIG. **3**, tier example, the EV switch **79** is constituted by a push switch disposed on the center console **771**, the priority mode is selected by pressing the EV switch **79** once if the EV priority mode is not selected, and the EV priority mode is canceled by pressing the EV switch **79** once if the EV priority mode is selected.
- (26) As shown in FIG. **1**, the travel control device **6** according to Embodiment 1 described above is connected to a multi-information display **8** for displaying various information. As shown in FIG. **6**,

the multi-information display **8** is provided with a display area **81** (indicator) for displaying that the EV priority mode is selected and a display area **83** displaying an operating state/non-operating state of the engine **25**.

(27) [Configuration of Travel Control Device **6** for Hybrid Vehicle **1**]

(28) As shown in FIG. **1**, the travel control device **6** for the hybrid vehicle **1** according to Embodiment 1 described above is provided with an acceleration control part **61** and a regenerative control part **63**. The acceleration control part **61** and the regenerative control part **63** are configured by causing the above-described processor to execute a control program stored in the memory. The acceleration control part **61** and the regenerative control part **63** control the traveling motor generator **21** and the power generation motor generator **23** via the power drive unit **3**, as well as control the engine **25** via the engine control unit **4**.

(29) The acceleration control part **61** controls the rotational drive of the traveling motor generator **21** via the power drive unit **3** as well as controls the rotational drive of the engine **25** via the engine control unit **4** such that the driving wheel **27** is power-driven based on the depression amount of the accelerator pedal detected by the accelerator pedal sensor **71**.

(30) For example, if the traveling motor generator **21** can output driving power necessary for the vehicle to travel with the electricity supplied from the driving battery **5**, the power drive unit **3** is controlled such that the traveling motor generator power-drives the driving wheel **27** of the vehicle in a state where the operation of the engine **25** is stopped. Whereby, the vehicle implements EV traveling.

(31) Further, for example, if the traveling motor generator **21** cannot output driving power necessary for the vehicle to travel (accelerate) with the electricity supplied from the driving battery **5**, such as in the case of rapid acceleration, and if the traveling motor generator **21** can output driving power necessary for the vehicle to travel (accelerate) by adding the electricity from the power generation motor generator **23**, the engine control unit **4** is controlled such that the engine **25** is started and the power drive unit **3** is controlled such that the power generation motor generator **23** generates electricity. The electricity generated by the power generation motor generator **23** is added to the electricity from the driving battery **5** and supplied to the traveling motor generator **21**. Further, for example, if the remaining battery capacity of the driving battery **5** is low, the engine control unit **4** is controlled such that the engine **25** is started and the power drive unit **3** is controlled such that the power generation motor generator **23** generates electricity. Electricity generated by the power generation motor generator **23** is charged to the driving battery **5** connected to the power drive unit **3**. Whereby, the vehicle implements series traveling.

(32) Further, for example, if the traveling motor generator **21** cannot output driving power necessary for the vehicle to travel (high speed travel) just by adding electricity from the power generation motor generator **23** to electricity supplied from the driving battery **5**, such as in a case of where the vehicle travels at high speed, the engine control unit **4** is controlled such that the engine **25** is started and the engine **25** power-drives the driving wheel **27** of the vehicle. Then, if the vehicle accelerates, the driving power from the traveling motor generator **21** is added to the driving power of the engine **25**. Whereby, the vehicle implements parallel traveling.

(33) The acceleration control part **61** according to Embodiment 1 includes an EV priority control portion **611**. The EV priority control portion **611** prioritizes EV traveling where the rotational drive of the engine **25** is stopped over engine traveling where the engine **25** is rotationally driven, if the EV priority mode is selected by the EV switch **79** and the EV priority control portion **611** selects the EV priority mode. Engine traveling according to Embodiment 1 is series traveling and parallel traveling described above, and the EV priority control portion **611** prioritizes EV traveling over series traveling and parallel traveling if the EV priority mode is selected by the EV switch **79**. For example, if the EV priority mode is selected, the EV priority control portion **611** increases the threshold of the depression amount of the accelerator pedal to switch from the EV traveling mode to the series traveling mode compared to that in a normal mode (if the EV priority mode is not

selected). Further, for example, if the EV priority mode is selected, the EV priority control portion **611** decreases the threshold of the remaining battery capacity of the driving battery **5** to switch from EV traveling to series traveling compared to that in the normal mode (if the EV priority mode is not selected). Furthermore, for example, if the EV priority mode is selected, the EV priority control portion **611** increases the threshold of the vehicle speed to switch from the EV traveling mode to the parallel traveling mode compared to that in the normal mode (if the EV priority mode is not selected).

(34) The above-described acceleration control part **61** switches from EV traveling to series traveling if the remaining battery capacity of the driving battery **5** falls below the threshold, if the temperature of the driving battery **5** extremely decreases, etc., and the acceleration control part **61** deselects the EV priority mode.

(35) The regenerative control part **63** controls the power drive unit **3** such that the traveling motor generator **21** performs regenerative braking on the driving wheel **27** at the regeneration level (regenerative braking intensity) set by the operation part **77**. For example, if the shift position is “D”, the regenerative control part **63** controls the power drive unit **3** such that the traveling motor generator **21** performs regenerative braking at the regenerative braking intensity “B2” when a foot is released from the accelerator pedal.

(36) The regenerative control part **63** includes a restriction portion **631**, a forced regeneration portion **633**, and a forced regeneration continuation portion **635**.

(37) The restriction portion **631** controls the power drive unit **3** such that the traveling motor generator **21** restricts or prohibits regenerative braking if the driving battery **5** (battery management unit **59**) restricts the acceptance of electricity to the driving battery **5**. Electricity generated in the traveling motor generator **21** is also restricted if the power drive unit **3** is controlled such that the traveling motor generator **21** restricts regenerative braking, and electricity generated in the traveling motor generator **21** is accepted to the driving battery **5** even if the acceptance of electricity to the driving battery **5** is restricted. No electricity is generated in the traveling motor generator **21** and no electricity is supplied to the driving battery **5** if the power drive unit **3** is controlled such that the traveling motor generator **21** prohibits regenerative braking.

(38) On the other hand, the restriction portion **631** lifts the restriction or prohibition of regenerative braking if the restriction on accepting electricity to the driving battery **5** is lifted in the state where the power drive unit **3** is controlled such that the traveling motor generator **21** restricts or prohibits regenerative braking. Whereby, the regenerative control part **63** controls the power drive unit **3** such that the traveling motor generator **21** performs regenerative braking on the driving wheel **27** at the regenerative braking intensity set by the operation part **77**, if the foot is released from the accelerator pedal.

(39) The forced regeneration portion **633** controls the power drive unit **3** such that the traveling motor generator **21** performs regenerative braking on the driving wheel **27** and controls the power drive unit **3** such that the power generation motor generator **23** is rotationally driven by electricity generated (regen rat-d) in the traveling motor generator **21** to rotationally drive the engine **25**, if the regenerative braking intensity is increased by the operation part **77** in the state where the power drive unit **3** is controlled such that the restriction portion **631** restricts or prohibits the regenerative braking of the traveling motor generator **21**. Controlling the power drive unit **3** such that the traveling motor generator **21** performs regenerative braking on the driving wheel **27** and controlling the power drive unit **3** such that the power generation motor generator **23** is rotationally driven by electricity generated in the traveling motor generator **21** to rotationally drive the engine **25** are referred to as “motoring control” in the following description. The motoring control is a control to consume electricity generated in the traveling motor generator **21**, and performing the motoring control makes it possible to control the power drive unit **3** such that the traveling motor generator **21** performs regenerative braking on the driving wheel **27** at the regenerative braking intensity set by the operation part **77** without collecting electricity to the driving battery **5**. Meanwhile, the

forced regeneration portion **633** discontinues the motoring control if the regenerative braking intensity is decreased by the operation part **77** in the state where the motoring control is performed. (40) In the motoring control, the power generation motor generator **23** is rotationally driven by electricity generated in the traveling motor generator **21**, and the engine **25** may be subjected to the combustion operation simultaneously via the engine control unit **4**. The combustion operation of the engine **25** performed simultaneously with the motoring control is controlled by the engine control unit **4** in terms of a fuel injection amount, a fuel injection timing, an air intake amount, and an ignition timing.

(41) The above-described acceleration control part **61** deselects the EV priority mode if the regenerative braking intensity is increased by the operation part **77** in the state where the EV priority mode is selected and the acceptance of electricity to the driving battery **5** is restricted. (42) Further, the above-described forced regeneration portion **633** transmits a cancellation notice of the EV priority mode to the acceleration control part **61** if the regenerative braking intensity is increased by the operation part **77**. Meanwhile, the acceleration control part **61** disables the selection of the EV priority mode from when the cancellation notice of the EV priority mode is received to when the driving battery **5** lifts the restriction on accepting electricity to the driving battery **5**.

(43) The forced regeneration continuation portion **635** continues the motoring control if the restriction on accepting electricity to the driving battery **5** is lifted in the state where the forced regeneration portion **633** performs the motoring control and during vehicle deceleration. Meanwhile, the forced regeneration continuation portion **635** discontinues the motoring control if the restriction on accepting electricity to the driving battery **5** is lifted in the state where forced regeneration motoring control is performed and other than during vehicle deceleration (for example, during acceleration). Whereby, the regenerative control part **63** controls the power drive unit **3** such that the traveling motor generator **21** performs regenerative braking on the driving wheel **27** at the regenerative braking intensity set by the operation part **77**, if engine noise is cut off and the foot is released from the accelerator pedal.

(44) The travel control device **6** for the hybrid vehicle **1** according to Embodiment 1 described above further includes a display control part **65**. The display control part **65** is a part for controlling display of the multi-information display **8**. For example, the display control part **65** displays that the EV priority mode is selected in the display area **81** of the multi-information display **8** for displaying that the EV priority mode is selected, if the EV priority mode is selected by the acceleration control part **61**. Further, for example, the display control part **65** displays that the engine **25** is in the operating state in the display area **83** of the multi-information display **8** for displaying the operating state/non-operating state of the engine **25**, if the engine **25** is operated, and the display control part **65** displays that the engine **25** is in the non-operating state in the display area **83** of the multi-information display **8** for displaying the operating state/non-operating state of the engine **25**, if the engine **25** is stopped.

(45) Further, for example, the display control part **65** displays that the EV priority mode is unselectable on the multi-information display **8**, if the acceleration control part **61** disables the selection of the EV priority mode. The display that the EV priority mode is unselectable may be, for example, a text display on the multi-information display **8**, or the display that the EV priority mode is unselectable may be illuminated.

(46) With the travel control device **6** for the hybrid vehicle **1** according to Embodiment 1 described above, the display control part **65** displays that the EV priority mode is selected in the display area **81** of the multi-information display **8** for displaying that the EV priority mode is selected, if the EV priority mode is selected by the acceleration control part **61**. Therefore, a user driving the hybrid vehicle **1** can know whether the EV priority mode is selected. Further, for example, the display control part **65** displays that the engine **25** is in the operating state in the display area **83** of the multi-information display **8** for displaying the operating state/non-operating state of the engine **25**,

if the engine 25 is operated, and the display control part 65 displays that the engine 25 is in the non-operating state in the display area 83 of the multi-information display 8 for displaying the operating state/non-operating state of the engine 25, if the engine 25 is stopped. Therefore, the user driving the hybrid vehicle 1 can know whether the engine 25 is in the operating state or in the non-operating state.

(47) Further, for example, the display control part 65 may display that the EV priority mode is unselectable, if the EV switch 79 is operated, from when the cancellation notice of the EV priority mode is received to when the driving battery 5 lifts the restriction on accepting electricity to the driving battery 5. Whereby, the user driving the hybrid vehicle 1 can know that the EV priority mode is unselectable from when the EV priority mode is canceled to when the driving battery 5 lifts the restriction on accepting electricity to the driving battery 5.

(48) [Contents of Travel Control of Hybrid Vehicle 1]

(49) As shown in FIG. 7, in the travel control device 6 for the hybrid vehicle 1 according to Embodiment 1, the EV priority control portion 611 first determines whether the EV priority mode is selectable (step S11). If the EV priority control portion 611 determines that the EV priority mode is selectable (step S11: Yes), the EV priority control portion 611 determines whether the EV switch 79 is pressed (step S12). If the EV priority control portion 611 determines that the EV switch 79 is pressed (step S12: Yes), the EV priority mode is selected and the display control part 65 displays that the mode is in the EV priority mode in the display area 81 of the multi-information display 8 for displaying the EV priority mode (step S13). On the other hand, if the EV priority control portion 611 determines that the EV priority mode is unselectable (step S11: No) or if the EV priority control portion 611 determines that the EV switch 79 is not pressed (step S12: No) even though the EV priority control portion 611 determines that the EV priority mode is selectable (step S11: Yes), the EV priority mode is not selected and the display control part 65 does not display that the mode is in the EV priority mode in the display area 81 of the multi-information display 8 for displaying the EV priority mode (step S14).

(50) Next, the travel control device 6 determines whether regenerative braking on the driving wheel 27 is possible (step S15). Whether regenerative braking on the driving wheel 27 is possible is determined by whether the traveling motor generator 21 is broken, regenerative braking on the driving wheel 27 is impossible if the traveling motor generator 21 is broken, and regenerative braking on the driving wheel 27 is possible if the traveling motor generator 21 is not broken.

(51) If the vehicle is not decelerated (if the vehicle is accelerated) (step S15: No), the acceleration control part 61 controls the traveling motor generator 21 and the power generation motor generator 23 via the power drive unit 3 as well as controls the engine 25 via the engine control unit 4 such that the driving wheel 27 is power-driven based on the depression amount of the accelerator pedal detected by the accelerator pedal sensor 71.

(52) The acceleration control part 61 switches from EV traveling to engine traveling as appropriate by using, for example, the depression amount and depression speed of the accelerator pedal detected by the accelerator pedal, the remaining battery capacity of the driving battery 5, and the vehicle speed as indexes. Then, if the EV priority mode is selected, EV traveling is prioritized over engine traveling as described above.

(53) If the vehicle is decelerated (step S15: Yes), the regenerative control part 63 refers to the determination of whether the battery management unit 59 can accept electricity to the driving battery 5 (step S16). Then, if the battery management unit 59 restricts the acceptance of electricity to the driving battery 5 (step S16: No), the restriction portion 631 controls the power drive unit 3 such that the traveling motor generator 21 restricts or prohibits regenerative braking (step S17).

(54) Next, if the regenerative braking intensity is increased by the operation part 77 (step S18: Yes) in the state where the restriction portion 631 controls the power drive unit 3 such that the traveling motor generator 21 restricts or prohibits regenerative braking on the driving wheel 27, the acceleration control part 61 deselects the EV priority mode and the forced regeneration portion 633

performs the motoring control (step S19). In Embodiment 1, the forced regeneration portion **633** controls the engine control unit **4** such that the engine **25** is subjected to the combustion operation simultaneously with the motoring control.

(55) Then, the forced regeneration portion **633** transmits the cancellation notice of the EV priority mode to the acceleration control part **611** (step S20). Whereby, the EV priority control portion **611** cancels the EV priority mode by receiving the cancellation notice of the EV priority mode, and the display control part **65** erases the display in the area of the multi-information display **8** for displaying the EV priority mode to disable the selection of the EV priority mode (step S21).

(56) Next, if the regenerative braking intensity is decreased by the operation part **77** (step S22: Yes) in the state where the motoring control is performed, the forced regeneration portion **633** discontinues the motoring control (step S23). In Embodiment 1, the forced regeneration portion **633** controls the engine control unit **4** such that the combustion operation of the engine **25** is discontinued simultaneously with the discontinuation of the motoring control. Whereby, the power drive unit **3** is controlled such that the restriction portion **631** restricts or prohibits regenerative braking of the traveling motor generator **21**, if engine noise is cut off and the foot is released from the accelerator pedal. If the regenerative braking intensity is not decreased by the operation part **77** (step S22: No) in the state where the motoring control is performed, the forced regeneration portion **633** continues the motoring control without discontinuing the motoring control. In Embodiment 1, the forced regeneration portion **633** controls the engine control unit **4** such that the motoring control is continued as well as the combustion operation of the engine **25** is continued.

(57) On the other hand, if the regenerative braking intensity is not increased by the operation part **77** (step S18: No) in step **18**, the restriction portion **631** continues the state where the power drive unit **3** is controlled such that the traveling motor generator **21** restricts or prohibits regenerative braking and enables the selection of the EV priority mode (Step S24).

(58) Further, if the battery management unit **59** does not restrict the acceptance of electricity to the driving battery **5** (step S16: Yes) in step **S16**, the regenerative control part **63** controls the power drive unit **3** such that the traveling motor generator **21** performs regenerative braking on the driving wheel **27** at the regeneration level (regenerative braking intensity) set by the operation part **77** (step S27) to enable the selection of the EV priority mode (step S28), on conditions that the motoring control is not performed (step S25: No) and the vehicle is not in deceleration (step S26: No).

(59) Thereafter, by repeating these steps, the travel control device **6** for the hybrid vehicle **1** power-drives the driving wheel **27** of the vehicle and performs regenerative braking on the driving wheel **27** of the vehicle.

(60) Therefore, if the acceptance of electricity to the driving battery **5** is restricted (step S16: No) and the restriction on accepting electricity to the driving battery **5** is lifted (step S19: Yes) when the forced regeneration portion **633** performs the motoring control (step S19), the forced regeneration continuation portion **635** discontinues the motoring control (step S27) on the condition the vehicle is not in deceleration (the vehicle is in acceleration) (step S26: No). Whereby, the regenerative control part **63** controls the power drive unit **3** such that the traveling motor generator **21** performs regenerative braking on the driving wheel **27** at the regenerative braking intensity set by the operation part **77**, if engine noise is cut off and the foot is released from the accelerator pedal.

(61) On the other hand, even if the acceptance of electricity to the driving battery **5** is restricted (step S16: No) and the restriction on accepting electricity to the driving battery **5** is lifted (step S16: Yes) when the forced regeneration portion **633** performs the motoring control (step S19: Yes), the forced regeneration continuation portion **635** continues the motoring control during vehicle deceleration (step S26: Yes). Further, in Embodiment 1, the forced regeneration portion **633** controls the engine control unit **4** such that the motoring control is continued as well as the combustion operation of the engine **25** is continued.

(62) With the travel control device **6** for the hybrid vehicle **1** according to Embodiment 1 described above, the acceleration control part **61** deselects the EV priority mode if the regenerative braking

intensity is increased by the operation part **77** in the state where the EV priority mode is selected and the acceptance of electricity to the driving battery **5** is restricted. Therefore, it is possible to avoid the rotational drive of the engine **25** by the power generation motor generator **23** in the state where the EV priority mode is selected.

(63) Further, the acceleration control part **61** disables the selection of the EV priority mode from when the regenerative braking intensity is increased by the operation part **77** to when the restriction on accepting electricity to the driving battery **5** is lifted in the state where the EV priority mode is selected and the acceptance of electricity to the driving battery **5** is restricted. Therefore, it is possible to maintain the state where the EV priority mode is canceled, from when the regenerative braking intensity is increased by the operation part **77** to when the driving battery **5** lifts the restriction on accepting electricity to the driving battery **5**.

(64) Furthermore, the motoring control is continued during vehicle deceleration even if the restriction on accepting electricity to the driving battery **5** is lifted while the motoring control is performed. Therefore, the discontinuation of the motoring control which is not due to a user operation disappears during vehicle deceleration, making it possible to suppress the risk of giving the user a strong feeling of strangeness, such as the possibility that the vehicle is broken.

(65) In the travel control device **6** for the hybrid vehicle **1** according to Embodiment 1 described above, the acceleration control part may select the EV priority mode (automatically return to the EV priority mode), if the driving battery **5** lifts the restriction on accepting electricity to the driving battery **5** after the EV priority mode is deselected. Consequently, there is no need to select the EV priority mode again by the EV switch **79**, even if the rotational drive of the power generation motor generator **23** is started by electricity generated in the traveling motor generator **21** and the EV priority mode is deselected in the case where the EV priority mode is selected by the EV switch **79**.

Embodiment 2

(66) The configuration of the hybrid vehicle **1** according to Embodiment 2 is not different from that of the hybrid vehicle **1** according to Embodiment 1, so the description is omitted.

(67) [Configuration of Travel Control Device **6** for Hybrid Vehicle **1**]

(68) The configuration of the travel control device **6** for the hybrid vehicle **1** according to Embodiment 2 is not different except for some configurations of the acceleration control part **61** and some configurations of the display control part **65**, so only different configurations will be described and descriptions of the same configurations will be omitted.

(69) In the travel control device **6** for the hybrid vehicle **1** according to Embodiment 2, the acceleration control part **61** selects the EV priority mode if the driving battery **5** lifts the restriction on accepting electricity to the driving battery **5** even though the rotational drive of the power generation motor generator **23** is started by electricity generated in the traveling motor generator in the state where the EV priority mode is selected and the EV priority mode is deselected. In other words, the acceleration control part **61** selects the EV priority mode (returns to the EV priority mode), if the driving battery **5** lifts the restriction on accepting electricity to the driving battery **5** after the EV priority mode is deselected.

(70) In the travel control device **6** for the hybrid vehicle **1** according to Embodiment 2, the display control part **65** displays that the EV priority mode is selected in the display area **81** of the multi-information display **8** for displaying that the EV priority mode is selected, if the EV priority mode is selected by the EV switch **79**. On the other hand, the display control part **65** erases the display that the EV priority mode is selected from the display area **81** of the multi-information display **8**, if the EV priority mode is deselected by the EV switch **79**. Whereby, if the EV priority mode is selected by the EV switch **79**, the display area **81** of the multi-information display **8** displays that EV priority mode is selected until the EV priority mode is canceled by the EV switch **79**.

(71) Further, in the travel control device **6** for the hybrid vehicle **1** according to Embodiment 2, the engine **25** may be started even if the display area **81** of the multi-information display **8** displays that the EV priority mode is selected. In such a case, since the feeling of strangeness may be given to

the user driving the hybrid vehicle **1**, the display control part **65** may displays that the engine **25** is started on the multi-information display **8** if the engine **25** is started.

(72) For example, the display control part **65** may display on the multi-information display **8** that the EV priority mode is selected (the mode automatically returns to the EV priority mode) after the driving battery **5** lifts the restriction on accepting electricity to the driving battery **5**, from when the EV priority mode is canceled (the cancellation notice of the EV priority mode is received) to when the restriction on accepting electricity is lifted. Whereby, the user driving the hybrid vehicle **1** can know that the EV priority mode is selected, i.e. that the mode automatically returns to the EV priority mode if the driving battery **5** lifts the restriction on accepting electricity to the driving battery **5**.

(73) [Contents of Travel Control of Hybrid Vehicle **1**]

(74) As shown in FIG. **8**, in the travel control device **6** for the hybrid vehicle **1** according to Embodiment 2, the EV priority control portion **611** first determines whether the mode is in the EV priority mode (step **S31**). If the EV priority control portion **611** determines that the mode is not in the EV priority mode (step **S31**: Yes), the EV priority control portion **611** determines whether the EV switch **79** is pressed (step **S32**). If the EV priority control portion **611** determines that the EV switch **79** is pressed (step **S32**: Yes), the EV priority mode is selected and the display control part **65** displays that the mode is in the EV priority mode in the display area **81** of the multi-information display **8** for displaying the EV priority mode (step **S33**). On the other hand, if the EV priority control portion **611** determines that the EV switch **79** is not pressed (step **S32**: No), the EV priority mode is not selected and the display control part **65** does not display that the mode is in the EV priority mode in the display area **81** of the multi-information display **8** for displaying the EV priority mode (step **S34**). Further, if the EV priority control portion **611** determines that the mode is in the EV priority mode (step **S31**: No) in step **S31**, the display control part **65** displays that the mode is in the EV priority mode in the display area **81** of the multi-information display **8** for displaying the EV priority mode (step **S33**).

(75) Next, the travel control device **6** determines whether regenerative braking on the driving wheel **27** is possible (step **S35**). Whether regenerative braking on the driving wheel **27** is possible is determined by whether the vehicle is decelerated, regenerative braking on the driving wheel **27** is impossible if the vehicle is not decelerated (if the vehicle is accelerated), and regenerative braking on the driving wheel **27** is possible if the vehicle is decelerated.

(76) If the vehicle is not decelerated (if the vehicle is accelerated) (step **S35**: No), the acceleration control part **61** controls the traveling motor generator **21** and the power generation motor generator **23** via the power drive unit **3** as well as controls the engine **25** via the engine control unit **4** such that the driving wheel **27** are power-driven based on the depression amount of the accelerator pedal detected by the accelerator pedal sensor **71**.

(77) The acceleration control part **61** switches from EV traveling to engine traveling as appropriate by using, for example, the depression amount and depression speed of the accelerator pedal detected by the accelerator pedal, the remaining battery capacity of the driving battery **5**, and the vehicle speed as the indexes. Then, if the EV priority mode is selected, EV traveling is prioritized over engine traveling as described above.

(78) If the vehicle is decelerated (step **S35**: Yes), the regenerative control part **63** refers to the determination of whether the battery management unit **59** can accept electricity to the driving battery **5** (step **S36**). Then, if the battery management unit **59** restricts the acceptance of electricity to the driving battery **5** (step **S36**: No), the restriction portion **631** controls the power drive unit **3** such that the traveling motor generator **21** restricts or prohibits regenerative braking (step **S37**).

(79) Next, if the regenerative braking intensity is increased by the operation part **77** (step **S38**: Yes) in the state where the restriction portion **631** controls the power drive unit **3** such that the traveling motor generator **21** restricts or prohibits regenerative braking on the driving wheel **27**, the acceleration control part **61** deselects the EV priority mode and the forced regeneration portion **633**

performs the motoring control (step S39). In Embodiment 2, the forced regeneration portion 633 controls the engine control unit 4 such that the engine 25 is subjected to the combustion operation simultaneously with the motoring control.

(80) Next, if the regenerative braking intensity is decreased by the operation part 77 (step S40: Yes) in the state where the motoring control is performed, the forced regeneration portion 633 discontinues the motoring control (step S41). In Embodiment 2, the forced regeneration portion 633 controls the engine control unit 4 such that the combustion operation of the engine 25 is discontinued simultaneously with the discontinuation of the motoring control. Whereby, the power drive unit 3 is controlled such that the restriction portion 631 restricts or prohibits regenerative braking of the traveling motor generator 21, if engine noise is cut off and the foot is released from the accelerator pedal. If the regenerative braking intensity is not decreased by the operation part 77 (step S40: No) in the state where the motoring control is performed, the forced regeneration portion 633 continues the motoring control without discontinuing the motoring control. In Embodiment 2, the forced regeneration portion 633 controls the engine control unit 4 such that the motoring control is continued as well as the combustion operation of the engine 25 is continued.

(81) On the other hand, if the regenerative braking intensity is not increased by the operation part 77 in step 38, the restriction portion 631 continues the state where the power drive unit 3 is controlled such that the traveling motor generator 21 restricts or prohibits regenerative braking.

(82) Further, if the battery management unit 59 does not restrict the acceptance of electricity to the driving battery 5 (step S36: Yes) in step S36, the regenerative control part 63 controls the power drive unit 3 such that the traveling motor generator 21 performs regenerative braking on the driving wheel 27 at the regeneration level (regenerative braking intensity) set by the operation part 77 (step S45), on conditions that the motoring control is not performed (step S42: No) and the vehicle is not in deceleration (step S43: No).

(83) Thereafter, by repeating these steps, the travel control device 6 for the hybrid vehicle 1 power-drives the driving wheel 27 of the vehicle and performs regenerative braking on the driving wheel 27 of the vehicle.

(84) Therefore, if the acceptance of electricity to the driving battery 5 is restricted (step S36: No) and the restriction on accepting electricity to the driving battery 5 is lifted (step S36: Yes) when the forced regeneration portion 633 performs the motoring control (step S39), the forced regeneration continuation portion 635 discontinues the motoring control (step S45) on a condition the vehicle is not in deceleration (the vehicle is in acceleration) (step S43: No). Whereby, the regenerative control part 63 controls the power drive unit 3 such that the traveling motor generator 21 performs regenerative braking on the driving wheel 27 at the regenerative braking intensity set by the operation part 77, if engine noise is cut off and the foot is released from the accelerator pedal.

(85) On the other hand, even if the acceptance of electricity to the driving battery 5 is restricted (step S36: No) and the restriction on accepting electricity to the driving battery 5 is lifted when the forced regeneration portion 633 performs the motoring control (step S39), the forced regeneration continuation portion 635 continues the motoring control during vehicle deceleration (step S43: Yes). Further, in Embodiment 2, the forced regeneration portion 633 controls the engine control unit 4 such that the motoring control is continued as well as the combustion operation of the engine 25 is continued.

(86) With the travel control device 6 for the hybrid vehicle 1 according to Embodiment 2 described above, the mode returns to the EV priority mode by causing the driving battery 5 to lift the restriction on accepting electricity to the driving battery 5 even if the rotational drive of the power generation motor generator 23 is started by electricity generated in the traveling motor generator 21 and the EV priority mode is deselected. Consequently, there is no need to select the EV priority mode again by the EV switch 79, even if the rotational drive of the power generation motor generator 23 is started by electricity generated in the traveling motor generator 21 and the EV priority mode is deselected in the case where the EV priority mode is selected by the EV switch 79.

(87) Further, the motoring control is continued during vehicle deceleration even if the restriction on accepting electricity to the driving battery 5 is lifted while the motoring control is performed. Therefore, the discontinuation of the motoring control which is not due to the user operation disappears during vehicle deceleration, making it possible to suppress the risk of giving the user the strong feeling of strangeness, such as the possibility that the vehicle is broken.

(88) The present invention is not limited to the above-described embodiments, and also includes an embodiment obtained by modifying the above-described embodiments and an embodiment obtained by combining these embodiments as appropriate.

(89) The contents described in the above embodiments would be understood as follows, for instance.

(90) [1] A travel control device (6) for a hybrid vehicle (1) according to an aspect, includes: a traveling motor generator (21) configured to power-drive a driving wheel (27) of the vehicle and configured to perform regenerative braking on the driving wheel (27); a driving battery (5) configured to supply electricity to the traveling motor generator (21) if the traveling motor generator (21) power-drives the driving wheel (27) and configured to recover electricity from the traveling motor generator (21) if the traveling motor generator (21) performs regenerative braking on the driving wheel (27); a power generation motor generator (23) configured to generate electricity to charge the driving battery (5); and an engine (25) configured to rotationally drive the power generation motor generator (23) and be rotationally driven by the power generation motor generator (23). The hybrid vehicle (1) includes: an EV switch (7)) for selecting an EV priority mode; and an operation part (77) for setting a regenerative braking intensity of the traveling motor generator (21). The travel control device (6) includes: an acceleration control part (61) configured to control rotational drive of the traveling motor generator (21) or rotational drive of the engine (25) such that the driving wheel (27) is power-driven based on a depression amount of an accelerator pedal; and a regenerative control part (63) configured to control the traveling motor generator (21) such that the traveling motor generator (21) performs regenerative braking on the driving wheel (27) with the regenerative control intensity set by the operation part (77). The acceleration control part (61) includes: an EV priority control portion (611) configured to prioritize EV traveling where the rotational drive of the engine (25) is stopped over engine traveling where the engine (25) is rotationally driven, if the EV priority mode is selected. The regenerative control part (63) includes: a restriction portion (631) configured to control the traveling motor generator (21) such that the regenerative braking is restricted or prohibited, if the driving battery (5) restricts acceptance of electricity to the driving battery (5); and a forced regeneration portion (633) configured to control the traveling motor generator (21) such that the traveling motor generator (21) performs regenerative braking on the driving wheel (27) and configured to rotationally drive the engine by electricity generated in the traveling motor generator (21) to consume electricity, if the regenerative braking intensity is increased by the operation part (77) in a state where the restriction portion (631) controls the traveling motor generator (21) such that the regenerative braking is restricted or prohibited. The acceleration control part (61) deselects the EV priority mode if the regenerative braking intensity is increased by the operation part (77) in a state where the EV priority mode is selected and the acceptance of electricity to the driving battery (7) is restricted.

(91) With such configuration, the acceleration control part (61) deselects the EV priority mode if the regenerative braking intensity is increased by the operation part (77) in the state where the EV priority mode is selected and the acceptance of electricity to the driving battery (7) is restricted. Therefore, it is possible to avoid the rotational drive of the engine (25) by the power generation motor generator (23) in the state where the EV priority mode is selected.

(92) [2] A travel control device (6) for a hybrid vehicle (1) according to another aspect is the travel control device (6) for the hybrid vehicle (1) as defined in [1] wherein the forced regeneration portion (633) transmits a cancellation notice of the EV priority mode to the acceleration control part (61) if the regenerative braking intensity is increased by the operation part (77), and wherein

the acceleration control part (61) disables selection of the EV priority mode from when the cancellation notice of the EV priority mode is received to when the driving battery (5) lifts the restriction on accepting electricity to the driving battery (5).

(93) With such configuration, the acceleration control part (61) disables the selection of the EV priority mode from when the regenerative braking intensity is increased by the operation part (77) to when the driving battery (5) lifts the restriction on accepting electricity to the driving battery (5). Whereby, it is possible to maintain the state where the EV priority mode is canceled, from when the regenerative braking intensity is increased by the operation part (77) to when the driving battery (5) lifts the restriction on accepting electricity to the driving battery (5).

(94) [3] A travel control device (6) for a hybrid vehicle (1) according to another aspect is the travel control device (6) for the hybrid vehicle (1) as defined in [1] or [2], wherein the acceleration control part (61) selects the EV priority mode, if the driving battery (5) lifts the restriction on accepting electricity to the driving battery (5) after the EV priority mode is deselected.

(95) With such configuration, the mode returns to the EV priority mode by causing the driving battery (5) to lift the restriction on accepting electricity to the driving battery (5) even if the rotational drive of the power generation motor generator (23) is started by electricity generated in the traveling motor generator (21) and the EV priority mode is deselected. Consequently, there is no need to select the EV priority mode again by the EV switch (79), even if the rotational drive of the power generation motor generator (23) is started by electricity generated in the traveling motor generator (21) and the EV priority mode is deselected in the case where the EV priority mode is selected by the EV switch (79).

(96) [4] A travel control device (6) for a hybrid vehicle (1) according to another aspect is the travel control device (6) for the hybrid vehicle (1) as defined in [2], wherein the hybrid vehicle (1) includes an indicator (the display area 81 of the multi-information display 8) for displaying that the EV priority mode is selected by the EV switch (79), and wherein the indicator (the display area 81 of the multi-information display 8) displays that the EV priority mode is unselectable, if the EV switch (79) is operated, from when the cancellation notice of the EV priority mode is received to when the driving battery (5) lifts the restriction on accepting electricity to the driving battery (5).

(97) With such configuration, the user driving the hybrid vehicle (1) can know that the EV priority mode is unselectable from when the EV priority mode is canceled to when the driving battery (5) lifts the restriction on accepting electricity to the driving battery (5).

(98) [5] A travel control device (6) for a hybrid vehicle (1) according to another aspect is the travel control device (6) for the hybrid vehicle (1) as defined in [2], wherein the hybrid vehicle (1) includes an indicator (the display area 81 of the multi-information display 8) for displaying that the EV priority mode is selected if the EV priority mode is selected, and wherein the indicator (the display area 81 of the multi-information display 8) displays that the EV priority mode is selected after the driving battery (5) lifts the restriction on accepting electricity to the driving battery (5), from when the cancellation notice of the EV priority mode is received to when the restriction is lifted.

(99) With such configuration, the user driving the hybrid vehicle (1) can know that the EV priority mode is selected if the driving battery (5) lifts the restriction on accepting electricity to the driving battery (5).

(100) [6] A travel control device (6) for a hybrid vehicle (1) according to an aspect is the travel control device (6) for the hybrid vehicle (1) as defined in any one of [1] to [5], including: a forced regeneration continuation portion (635) configured to continue a control to rotationally drive the power generation motor generator (23) by electricity generated in the traveling motor generator (21), during deceleration of the hybrid vehicle (1), even if the restriction on accepting electricity to the driving battery (5) is lifted when the forced regeneration portion (633) controls the traveling motor generator (21) such that the traveling motor generator (21) performs regenerative braking on the driving wheel (27).

(101) With such configuration, the forced regeneration continuation portion (635) continues the control to rotationally drive the power generation motor generator (23) by electricity generated in the traveling motor generator (21), during deceleration of the hybrid vehicle (1), even if the restriction on accepting electricity to the driving battery (5) is lifted when the forced regeneration portion (633) controls the traveling motor generator (21) such that the traveling motor generator (21) performs regenerative braking on the driving wheel (27). Whereby, the state is continued where the engine (225) is rotationally driven even if the restriction on accepting electricity to the driving battery (5) is lifted during deceleration of the hybrid vehicle (1). Therefore, the stop of the engine which is not due to a user operation disappears during vehicle deceleration, making it possible to suppress the risk of giving the user a strong feeling of strangeness, such as the possibility that the vehicle is broken.

REFERENCE SIGNS LIST

(102) **1** Hybrid vehicle **21** Traveling motor generator **23** Power generation motor generator **25** Engine **27** Driving wheel **29** Transaxle **291** Clutch **3** Power drive unit (PDU) **4** Engine control unit (ECU) **5** Driving battery **50** Battery pack **51** Battery cell **52** Battery module **53** Current sensor **55** Leakage sensor **57** Cell monitor unit (CMU) **59** Battery management unit (BMU) **6** Travel control device (EVECU) **61** Acceleration control part **611** EV priority control portion **63** Regenerative control part **631** Restriction portion **633** Forced regeneration portion **635** Forced regeneration continuation portion **65** Display control part **71** Accelerator pedal sensor **73** Brake pedal sensor **75** Vehicle speed sensor **77** Operation part **771** Center console **773** Selector lever **775** Steering **777** Regeneration level selector **7771** Negative-side selector **7773** Positive-side selector **79** EV switch **8** Multi-information display (indicator) **81** Display area for displaying EV priority mode **83** Display area for displaying operating state/non-operating state of engine

Claims

1. A travel control device for a hybrid vehicle, comprising: a traveling motor generator configured to power-drive a driving wheel of the vehicle and configured to perform regenerative braking on the driving wheel; a driving battery configured to supply electricity to the traveling motor generator if the traveling motor generator power-drives the driving wheel and configured to recover electricity from the traveling motor generator if the traveling motor generator performs regenerative braking on the driving wheel; a power generation motor generator configured to generate electricity to charge the driving battery; and an engine configured to rotationally drive the power generation motor generator and be rotationally driven by the power generation motor generator, wherein the hybrid vehicle comprises: an EV switch for selecting an EV priority mode; and an operation part for setting a regenerative braking intensity of the traveling motor generator, wherein the travel control device comprises: an acceleration control part configured to control rotational drive of the traveling motor generator or rotational drive of the engine such that the driving wheel is power-driven based on a depression amount of an accelerator pedal; and a regenerative control part configured to control the traveling motor generator such that the traveling motor generator performs regenerative braking on the driving wheel with the regenerative control intensity set by the operation part, wherein the acceleration control part includes: an EV priority control portion configured to prioritize EV traveling where the rotational drive of the engine is stopped over engine traveling where the engine is rotationally driven, if the EV priority mode is selected, wherein the regenerative control part includes: a restriction portion configured to control the traveling motor generator such that the regenerative braking is restricted or prohibited, if the driving battery restricts acceptance of electricity to the driving battery; and a forced regeneration portion configured to control the traveling motor generator such that the traveling motor generator performs regenerative braking on the driving wheel and configured to rotationally drive the power generation motor generator by electricity generated in the traveling motor generator to rotationally

drive the engine, if the regenerative braking intensity is increased by the operation part in a state where the restriction portion controls the traveling motor generator such that the regenerative braking is restricted or prohibited, and wherein the acceleration control part deselects the EV priority mode if the regenerative braking intensity is increased by the operation part in a state where the EV priority mode is selected and the acceptance of electricity to the driving battery is restricted.

2. The travel control device for the hybrid vehicle according to claim 1, wherein the forced regeneration portion transmits a cancellation notice of the EV priority mode to the acceleration control part if the regenerative braking intensity is increased by the operation part, and wherein the acceleration control part disables selection of the EV priority mode from when the cancellation notice of the EV priority mode is received to when the driving battery lifts the restriction on accepting electricity to the driving battery.

3. The travel control device for the hybrid vehicle according to claim 2, wherein the hybrid vehicle comprises an indicator for displaying that the EV priority mode is selected by the EV switch, and wherein the indicator displays that the EV priority mode is unselectable, if the EV switch is operated, from when the cancellation notice of the EV priority mode is received to when the driving battery lifts the restriction on accepting electricity to the driving battery.

4. The travel control device for the hybrid vehicle according to claim 2, wherein the hybrid vehicle comprises an indicator for displaying that the EV priority mode is selected if the EV priority mode is selected, and wherein the indicator displays that the EV priority mode is selected after the driving battery lifts the restriction on accepting electricity to the driving battery, from when the cancellation notice of the EV priority mode is received to when the restriction is lifted.

5. The travel control device for the hybrid vehicle according to claim 1, wherein the acceleration control part selects the EV priority mode, if the driving battery lifts the restriction on accepting electricity to the driving battery after the EV priority mode is deselected.

6. The travel control device for the hybrid vehicle according to claim 1, comprising: a forced regeneration continuation portion configured to continue a control to rotationally drive the power generation motor generator by electricity generated in the traveling motor generator, during deceleration of the hybrid vehicle, even if the restriction on accepting electricity to the driving battery is lifted when the forced regeneration portion controls the traveling motor generator such that the traveling motor generator performs regenerative braking on the driving wheel.
